L-P-398B
May 30, 1973
SUPERSEDING
Fed. Spec. L-P-398A
June 24, 1968

FEDERAL SPECIFICATION

PLASTIC MOLDING MATERIAL, STYRENE-BUTADIENE

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers styrene-butadiene molding and extrusion material (see 6.1).

1.2 Classification.

1.2.1 Types and grades. Styrene-butadiene molding and extrusion material shall be of the following types and grades, as specified (see 3.1, 6.1, and 6.2):

Type I - Lowest impact strength
Type II - Moderate impact strength
Type III - Higher impact strength
Type IV - Highest impact strength
Grade A - Lower Vicat softening point
Grade B - Higher Vicat softening point

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

/FSC 9330/

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Federal Specifications:

PPP-D-723 - Drums, Fiber.
PPP-D-729 - Drums, Shipping and Storage, Steel, 55-gallon.

Federal Standard:

Fed. Std. No. 123 - Marking for Domestic Shipments (Civil Agencies).

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington, D. C. 20402.)

(Single copies of this specification and other Federal specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Washington, D. C., Atlanta, Chicago, Kansas City, Mo., Fort Worth, Denver, San Francisco, Los Angeles and Seattle, Washington.)

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from the established distribution points in their agencies.)

Military Standards:

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

(Copies of Military Specifications and Standards, required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

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American Society for Testing and Materials (ASTM) Standards:

D 256-72A	- Impact Resistance of Plastics and Electrical Insulating Materials.
D 618-61 (1971)	- Conditioning Plastics and Electrical insulating Materials for Testing.
D 621-64	- Deformation of Plastics Under Load.
D 638-68	- Tensile Properties of Plastics.
D 785-65 (1970)	- Rockwell Hardness of Plastics and Electrical Insulating Materials.
D 792-66 (1970)	- Specific Gravity and Density of Plastics by Displacement.
D 1897-68	- Inject Molding of Specimens of Polystyrene Molding and Extrusion Materials.
D 2292-68	- Compression Molding of Specimens of Styrene Butadiene Molding and Extrusion Materials.
D 1525-70	- Vicat Softening Point of Plastics.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103.)

National Motor Freight Traffic Association, Incorporated, Agent:

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., Tariff Order Section, 1616 P Street N.W., Washington, D. C. 20036.)

Uniform Classification Committee, Agent:

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, Illinois 60606).

(Technical Society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

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3. REQUIREMENTS

3.1 Material. The material shall consist of styrene butadiene copolymers or a blend of polymers or copolymers of styrene and butadiene, with or without the addition of colorants and small amounts of lubricants.

3.2 Property values. When tested as specified in the applicable procedure of 4.3, the material shall conform to the property values shown in table I.

3.3 Form. The material shall be furnished in the form of molding powder, granules, or pellets, as specified (see 6.2). When a particular particle size is specified, the limits and method of testing shall be as specified by the procuring agency (see 6.5).

3.4 Color. The color and opacity of the material, as determined on plaques molded from it, shall be as specified by the procuring agency (see 6.2).

3.5 Suitability for use with explosives and chemicals. When suitability for use with a particular explosive or chemical is specified, the material shall be approved by the procuring agency. The suitability shall be verified in a Government laboratory and the method of inspection shall be as specified by the procuring agency (see 6.2 and 6.4).

3.6 Health hazard. The material shall have no adverse effect on the health of personnel when used for its intended purpose. Evidence to this effect shall be subject to review by departmental medical authority (see 6.6).

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Table I. Property values

Property	Type I		Type II		Type III		Type IV
	Grade A	Grade B	Grade A	Grade B	Grade A	Grade B	Grade A
Specific gravity at 23°/23°C. (73.4°/73.4°F.):[1,2]							
Max.	1.09	1.09	1.09	1.09	1.09	1.09	1.09
Min.	1.04	1.04	1.04	1.04	1.04	1.04	1.04
Tensile yield stress, psi, min.							
Compression moldings	2000	5000	1500	3500	1500	2400	1000
Injection moldings	3500	6500	3000	5000	3000	3900	2500
Elongation, percent[2], min.	5.0	3.0	15.0	5.0	20.0	20.0	20.0
Rockwell hardness, L scale, R. H. number[2], min.	80	80	60	60	5	5	-5
Impact strength, Izod, foot pound per inch of notch, 1/8 by 1/2 by 2-1/2 inch specimen[3], min.	0.35	0.35	0.60	0.60	1.00	1.00	3.00
Vicat softening point, min. °F.	176	203	176	203	176	203	176
Deformation under load at 2000 p.s.i. and 50° ± 1°C. (122° ± 1.8°F.) percent[1,2], max.	2.5	1.5	7.0	5.0	15.0	10.0	15.0

[1]For uncolored, unpigmented, unfilled materials.
[2]Use injection molded test specimens which have not been annealed.
[3]Use compression molded test specimens.

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3.7 Workmanship. The powder, granules or pellets shall be uniform in color within each container and from container to container. The material shall be uniform in form, from container to container, and shall be free from contamination as determined by visual examination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated. For purposes of sampling, an inspection lot for examination and tests shall consist of all material of the same type and grade submitted for delivery at one time.

4.2.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified, or if none, in accordance with this specification. In the event of conflict, this specification shall govern. A supplier's certificate of compliance with 3.1 shall be furnished.

4.2.2 Inspection of Material.

4.2.2.1 Examination of the material. Examination of the material shall be made in accordance with the list of defects, inspection levels and acceptable quality levels (AQLs) set forth below. The lot size, for purpose of determining the sample size in accordance with MIL-STD-105, shall be expressed in units of 100 pounds for examination in 4.2.2.1.1, and in units of shipping containers for examination in 4.2.2.1.2.

4.2.2.1.1 Examination of the material for defects in appearance and workmanship. The sample unit for this examination, specified in table II, shall be approximately one pound.

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Table II. Examination of the material for defects in appearance and workmanship

Examine	Defect
Appearance and workmanship	Form or color not uniform Form not as specified Color not as specified Not clean, presence of foreign material or other contamination

4.2.2.1.2 Examination of the preparation for delivery requirements. An examination shall be made in accordance with Table III to determine that packaging, packing and marking comply with requirements specified in section 5. The sample unit for this examination shall be one shipping container fully packed, selected just prior to the closing operation. Shipping containers fully prepared for delivery shall be examined for closure defects.

Table III. Examination of preparation for delivery

Examine	Defect
Packing	Not level specified; not in accordance with contract requirements Any nonconforming component, component missing, damaged or otherwise defective affecting serviceability Inadequate application of components such as: Incomplete closures of case liners; container flaps, loose or inadequate strappings, bulged or distorted containers
Quantity of material	Less than specified or indicated quantity
Weight	Gross weight exceeds specified requirements
Markings	Interior or exterior markings omitted, illegible, incorrect, incomplete, of improper size, location sequence, method of application, or not in accordance with contract requirements.

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4.2.2.1.3 Inspection levels and acceptable quality levels (AQLs) for examinations. The inspection levels for determining the sample size and the acceptable quality level (AQL) expressed as defects per 100 units shall be as follows:

Examination paragraph	Inspection level	AQL
4.2.2.1.1	II	2.5
4.2.2.1.2	S-2	2.5

4.2.3 Testing. The material shall be tested for the applicable characteristics listed in table I in accordance with the test methods specified herein for each lot submitted for inspection. The lot size, for the purpose of determining the sample size for testing shall be expressed in units of 100 pounds. The sample unit shall consist of sufficient material to prepare all required specimens. The inspection level shall be S-1 with an acceptance number of 0. The results for each test shall be the averaged results of the specimens, except where only one specimen is to be tested.

4.2.3.1 Classification of tests. All tests shall be classified as follows:

(a) Lot acceptance tests (see 4.2.3.2).
(b) Periodic lot check tests (see 4.2.3.3).

4.2.3.2 Lot acceptance tests. Lot acceptance tests shall be made on each lot of material and shall be the basis for acceptance or rejection of the lot, except when periodic lot check tests are required. The lot acceptance tests shall consist of the tests shown in table IV.

Table IV. Lot acceptance tests

Property	Applicable to type	Reference Paragraph
Tensile yield strength	All types	4.3.4
Impact strength	Types II, III and IV	4.3.6
Vicat softening point	All types	4.3.7

4.2.3.3 Periodic lot check tests. Periodic lot check tests shall be made on the first lot of material furnished under this specification, and on any subsequent lot specified by the procuring agency (see 6.2). Periodic lot check tests shall consist of all the tests specified in table I. If specified by the procuring agency, tests for form and color shall be included as periodic lot check tests. When periodic lot check tests are made, they shall be included in the basis for acceptance or rejection of the lot.

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4.3 Test methods.

4.3.1 Specimen preparation. Unless otherwise specified, all specimens except that for impact strength testing shall be prepared by injection molding. Injection molding shall be in accordance with ASTM D 1897-68. Specimens for impact testing shall be prepared by compression molding. Compression molding shall be in accordance with ASTM D 2292-68. An option is given to use either injection or compression molded specimens for tensile yield stress determinations.

4.3.2 Specimen conditioning and testing. All specimens shall be conditioned in accordance with procedure A of ASTM D 618-61 (1971) except that the minimum conditioning time shall be 16 hours. Specimens for determining deformation under load shall be conditioned in accordance with procedure B of ASTM D 618-61 (1971). Testing shall be at 23° ± 1° C. (73.4° ± 1.8° F.) and 50 ± 5 percent relative humidity, except for deformation under load determinations (see 4.3.8).

4.3.3 Specific gravity. One specimen shall be tested in accordance with method A of ASTM D 792-66 (1970).

4.3.4 Tensile yield strength and elongation. Three specimens, each 1/8 ± 1/100 inch thick shall be tested in accordance with ASTM D 638-68, using testing speed B. Dimensions of each specimen shall conform to type I of figure 1 of ASTM D 638-68. Test results shall include whether injection or compression molded specimens were used.

4.3.5 Hardness. Three specimens shall be tested in accordance with method A of ASTM D 785-65 (1970)

4.3.6 Impact strength. Three specimens shall be tested in accordance with method A of ASTM D 256-72a, except that the notch shall be produced by a machining tool using a single-tooth type of milling cutter. Specimens shall be 1/8 by 1/2 inch bar.

4.3.7 Vicat softening point. Three specimens shall be tested in accordance with ASTM D 1525-70, using rate B.

4.3.8 Deformation under load. Three specimens shall be tested in accordance with method A of ASTM D 621-64. The temperature of testing shall be 50° ± 1.0° C. (122° ± 1.8° F.).

5. PREPARATION FOR DELIVERY

5.1 Packing. Packing shall be level A, B, or C, as specified (see 6.2).

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5.1.1 Level A. Unless otherwise specified (see 6.2), the material in the quantity specified (see 6.2), shall be packed in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type II, grade A, or type III, grade A in quantities of 200 pounds maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds maximum.

Insofar as practical, containers shall be of uniform shape and size, with minimum cube and tare consistent with the protection required. Containers shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with an 0.004 inch thick polyethylene liner properly heat sealed.

5.1.2 Level B. Unless otherwise specified (see 6.2), the material, in the quantity specified (see 6.2), shall be in one of the following types of containers:

(a) Fiber drums conforming to PPP-D-723, type I, grade A in quantities of 200 pounds maximum.

(b) Metal drums conforming to PPP-D-729, type III or type IV, in quantities of 400 pounds maximum.

Insofar as practical, containers shall be of uniform shape and size with minimum cube and tare consistent with the protection required. Containers shall contain identical quantities and shall be closed in accordance with the applicable container specification. Fiber drums shall be furnished with an 0.004 inch thick polyethylene liner properly heat sealed.

5.1.3 Level C. Packing shall be in a manner to insure carrier acceptance and safe delivery at destination at the lowest transportation rate for such supplies. Containers shall be in accordance with Uniform Freight Classification Rules or National Motor Freight Classification Rules, as applicable.

5.2 Marking.

5.2.1 Civil agencies. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with Fed. Std. No. 123.

5.2.2 Military requirements. In addition to any special marking required by the contract or order, shipping containers shall be marked in accordance with MIL-STD-129.

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6. NOTES

6.1 Intended use. This material is not intended for applications where most plastic materials may become health hazards (see 6.6). Types I and II are intended for general purpose applications requiring greater toughness, than is offered by straight polystyrene, and the inert properties of this material. Types III and IV are intended for applications which require greater elongation and impact strength than can be obtained with type I and II materials. Styrene-butadiene materials are intended for use in ammunition components, also.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following data in procurement documents:

(a) Title, number and date of this specification.
(b) Type and grade of material required (see 1.2).
(c) Form, if required (see 3.3), and method of testing when applicable.
(d) Color, and opacity, if required (see 3.4).
(e) Suitability for use with explosives and chemicals, if required (see 3.5).
(f) Periodic lot check tests when required (see 4.2.3.3).
(g) Selection of level of packing required (see 5.1).
(h) If other containers are required (see 5.1.1 and 5.1.2).
(i) Quantity in each shipping container (see 5.1.1 and 5.1.2).
(j) Special markings if required (see 5.2.1 and 5.2.2).

6.3 The mechanical property requirements of this specification are those which are considered adequate for assuring the procurement of uniform high quality materials. The values specified should not be used as design data where specimen or part geometry, enviroment, or rate or method of load application, may vary from the standard test conditions.

6.4 Suitability for use with explosives. Information concerning suitability of many plastics for use with various explosives and chemicals under various conditions is on file at Picatinny Arsenal, Dover, New Jersey. Procuring activities desiring information on this subject should first contact Picatinny Arsenal to determine whether the information is already available.

6.5 Fabricator choice of form. The fabricator of molded parts or extruded shapes is expected to specify the material form best suited to his equipment, (see 3.3).

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6.6 Questions raised regarding toxicity should be referred by the procuring agency to the departmental medical authority. In the case of Army procurement, the Surgeon General will act as advisor to the procuring agency.

MILITARY CUSTODIANS:

Army - MR
Air Force - 11

Review activities:

Army - MU, EL, GL
Navy - SH
Air Force - 84

User activities:

Army - ME
Navy - OS, AS

Preparing activity:

Army - MR

CIVIL AGENCIES INTEREST:

GSA-FSS
HEW-NIH
VA-DMS
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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

L-P-398B, Plastic Molding Material, Styrene-Butadiene

ORGANIZATION	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 886? WHICH IS OBSOLETE

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